

COEFFICIENT OF RESTITUTION

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Work on file paper

1. Sphere A, of mass 3 kg, and Sphere B, of mass 5 kg, are travelling towards each other along a straight line. They collide when A has speed 7 ms^{-1} and B has speed 2 ms^{-1} . After the impact sphere B is travelling in the opposite direction at a speed of 4 ms^{-1} . Find:
 - a. the speed and direction of A after the collision;
 - b. the coefficient of restitution for this collision;
2. Spheres A and B, each of mass 0.5 kg, collide when travelling in the same direction along the same straight line. Before the collision A has speed 3 ms^{-1} and B has speed 2 ms^{-1} . The collision is perfectly elastic (coefficient of restitution $e = 1$).
 - a. Find the speed and direction of both spheres after the collision.
 - b. Show that there is no loss of kinetic energy in this collision. (Kinetic energy $= \frac{1}{2}mv^2$)
3. Spheres A and B of equal mass collide when travelling with velocities u_A and u_B respectively. If the collision is perfectly elastic, show that after impact the velocities of A and B are u_B and u_A respectively.
4. Spheres A and B, each of mass 2 kg, collide when travelling towards each other along a straight line. Before the collision sphere A has speed 1 ms^{-1} and sphere B has speed 5 ms^{-1} . If the coefficient of restitution for the collision is $\frac{1}{3}$, find:
 - a. the speed of both spheres after the collision;
 - b. the loss of kinetic energy caused by the collision.
5. A particle of mass $5m$, moving in a straight line with velocity u explodes, splitting into two fragments: A of mass $2m$, and B of mass $3m$. Fragment B moves with velocity $3u$ in the same direction as the original particle.
 - a. Explain why the combined momentum of the two fragments will be the same as that of the original particle.
 - b. Find the velocity of A. State the direction in which A is traveling.
 - c. Find the kinetic energy generated by the explosion.
6. Two spheres of masses m kg and $3m$ kg collide when they are moving in a straight line with speeds $2u$ and u respectively. The coefficient of restitution between them is $\frac{1}{5}$. Find their speeds after impact if they collide when moving:
 - a. in the same direction;
 - b. in opposite directions.
7. Two spheres of equal mass collide head-on when moving with speeds u and $2u$. Half of the original kinetic energy is lost on impact. Find the coefficient of restitution between the spheres.

Bonus questions:

8. Two smooth spheres A and B of masses m_A and m_B respectively collide when moving with velocities u_A and u_B respectively. The coefficient of restitution between the spheres is e .
 - a. Find the velocities of the spheres after the impact.
 - b. Show that the loss in kinetic energy caused by the impact is

$$\frac{1}{2} \left(\frac{m_A m_B}{m_A + m_B} \right) (u_A - u_B)^2 (1 - e^2).$$

- c. What happens if the collision is perfectly elastic?
- d. Explain why the value of the coefficient of restitution in a collision such as the one described above is usually at most 1. Under what circumstances might it be greater than 1?

9. Three smooth spheres A , B , and C , of masses $3m$, $2m$, and m respectively, are lying in a straight line on a smooth horizontal plane. A is projected with speed u to collide directly with B which goes on to collide directly with C . The coefficient of restitution between each pair of spheres is $\frac{3}{4}$.
- Show that after the first impact the velocities of A and B are $\frac{3}{10}u$ and $\frac{21}{20}u$ respectively
 - Show that after the second impact the velocities of A , B , and C are $\frac{3}{10}u$, $\frac{7}{16}u$ and $\frac{49}{40}u$ respectively.
 - Explain why there will be no further impacts.